Sri Vishnu Sathvik Peddinti

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.74
Sri Gayatri Educational Institutions	Class XII (Telangana State Board Of Intermediate Education)	2019	84.20%
Vikas The Concept School	Class X (Central Board Of Secondary Education)	2017	8.40%

WORK EXPERIENCE

Research Assistant | TiHAN (Technology Innovation Hub on Autonomous Navigation)

- Contributed to successfully deploying the **Autonomous Campus Shuttle (ACS)** at the IIT Hyderabad campus.
- Revamped LiDAR-based navigation code for the ACS through **re-parameterisation and modularisation,** while automating **parameter tuning and sensor calibration** to minimise manual intervention and streamline autonomous vehicle deployments.
- Integral member of a two-person team responsible for **developing and integrating** key features for the ACS, including **Adaptive Speed Control, GNSS-LiDAR fusion, LiDAR-Camera fusion, Automatic Sensor Calibration,** and **Sensor Fusion-based Obstacle Avoidance**.
- **Collaborated** closely with senior team members and mentored juniors in conducting numerous **demonstrations** and successful deployments of the ACS at various high-profile events and locations.
- Proactively devised solutions for challenges assigned, demonstrating **adaptability** and **problem-solving skills**.

Project Assistant | Department of Electrical Engineering, IIT Hyderabad [Link]

- Contributed to **LiDAR-based road area detection** using intensity information.
- Implemented a **LiDAR-based Dynamic Window Approach (DWA)** for local path planning and autonomous navigation.
- Implemented **depth-camera-based object detection and distance estimation** for real-time obstacle perception.

Skills: Computer Vision, Sensor Fusion, ROS, Deep Learning, Algorithm Development, Team Management, Communication.

PROJECTS

M.Tech Thesis Project | Multi-Modal Perception and Sensor Fusion for Autonomous Navigation using LiDAR and Camera

- Developed a **LiDAR-Radar fusion framework** for enhanced obstacle detection, velocity estimation, and tracking, enabling smooth and robust obstacle avoidance.
- Developed an **auto-calibration pipeline** using ICP to estimate **extrinsic transformation parameters** between two LiDARs for accurate point-cloud alignment and sensor fusion.
- Designed a **Dynamic Region of Interest (D-ROI)** leveraging LiDAR-based obstacle clustering and filtering to improve perception and enable robust autonomous navigation.

Geo-Referencing Dense FARO Indoor Maps Using GNSS-NDT Correspondences for Absolute Indoor Positioning [github]

- Developed a **GNSS-NDT-based geo-referencing pipeline** to align dense FARO indoor RGBXYZ point-cloud maps with the global **ENU/WGS-84 coordinate frame.**

- Estimated a **3D similarity transformation using the Umeyama method** from sparse, non-collinear GNSS-NDT correspondences to globally anchor the complete indoor map.
- Enabled **absolute indoor positioning in GNSS-denied environments**, achieving **0.134 m RMSE** using only three outdoor GNSS-NDT correspondences.

Real-Time LiDAR Map-Based NDT Localisation Failure Detection and Automatic Relocalisation [github]

- Developed a **real-time localisation watchdog** using **LiDAR pose, IMU yaw consistency, velocity, acceleration, and localisation execution-time monitoring** to detect localisation failures.
- Implemented an **automatic recovery mechanism** that stores the **last valid LiDAR pose**, triggers navigation stop on failure, and reinitialises localisation on the **pre-built LiDAR map** using adaptive pose offsets.
- Designed a localisation state machine **and recovery validation** with pose stability checks, timeout monitoring, and repeated reinitialisation, enabling **automatic navigation resumption** after successful relocalisation.

Sensor Fusion of LiDAR and Camera for Depth-Enhanced Object Detection [github]

- Developed a **LiDAR-camera fusion framework** to associate camera-based object detections with **3D LiDAR depth**, enabling distance-aware obstacle perception.
- Integrated **camera-based road segmentation with LiDAR point clouds** through calibrated 3D-to-2D projection, achieving **96% drivable-area and 95% lane-detection accuracy** using TwinLiteNet.
- Used **LiDAR intensity/retroreflectivity and camera semantic information** to selectively filter relevant points for traffic sign, **obstacle, and road detection**, providing reliable perception inputs for navigation and path planning.

PUBLICATIONS

- "Geo-Referencing Dense FARO Indoor Maps Using GNSS-NDT Correspondences for Absolute Indoor Positioning" S. Sathvik, Parvez Alam, P. Rajalakshmi, IEEE ICRAE 2026, Nagoya, Japan

- "Aligning LiDAR to Vision Language Space: A Fusion Approach for Steering Angle Regression of a Vehicle", Alam Parvez, Sathvik V., P. Rajalakshmi, IEEE IJCNN Workshop Proceedings, 2025

- "LiDAR-GNSS Fusion to Initiate Localisation at Intermediate Points on a 3D Point Cloud Map", A. Thakur, C. A. R. Ram, S. Sathvik, et.al. IEEE Vehicular Technology Conference Fall, Washington, 2024

- "Autonomous Cooperative Platooning Powered by LiDAR-Guided Adaptive Cruise Control", A. Thakur, C. A. R. Ram, S. Sathvik, B. Badugu, et.al. IEEE Vehicular Technology Conference Spring, Singapore, 2024

PATENTS

- "System And Method For Planning A Trajectory For An Autonomous Ground Vehicle" applied on 26th September 2025, Application Number:202541092386

SKILLS

- **Programming Languages:** Python, C, ROS, MATLAB
- **Technologies:** SLAM, LIDAR and Camera data processing, Sensor calibration, Sensor fusion, Path planning, Computer vision, Machine learning, Deep learning, Pure pursuit, MPC, NDT localization, A-LOAM mapping.
- **Libraries**: Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, PyTorch, XGBoost, OpenCV.
- **Sensors:** Velodyne VLP-16, VLP-32, HDL-64, VLS-128 LiDAR, LiVOX HAP LiDAR, LiVOX MID 360, FARO TLS, Ouster OS2-128 LiDAR, Robosense Ruby Plus 128-channel LiDAR, Robosense M1, Zed Stereo Camera, Intel Realsense stereo camera, Basler Camera and Novatel GNSS/INS.
- **Operating Systems:** Windows, Linux.

POSITION OF RESPONSIBILITY

Teaching Assistant - Autonomous Navigation (Course Code: EE50013)

- Teaching Assistant for the **Autonomous Navigation course (EE50013)** during the **January-June 2026 semester**.

RELEVANT COURSES

Matrix Theory || Random Variables and Stochastic Processes || Principles of Digital Communication || || Convex

Optimisation || Autonomous Navigation || Wireless Sensor Network || DSP & FPGA Lab || Automotive Communication and Sensing || Image and Video Processing ||

HOBBIES AND INTRESTS

- **Rapping** - Passionate about rap music and have performed live during B.Tech and at IIT Hyderabad campus events.
- **Racing & Karting** - Strong interest in motorsports, with a particular passion for **racing and karting**; actively follow **Formula 1**, especially the **Scuderia Ferrari** team.
- **Travelling** - Enjoy exploring new places, experiencing different environments, and discovering new cultures.
- **Music** - Strong interest in music and actively engage in various musical activities.